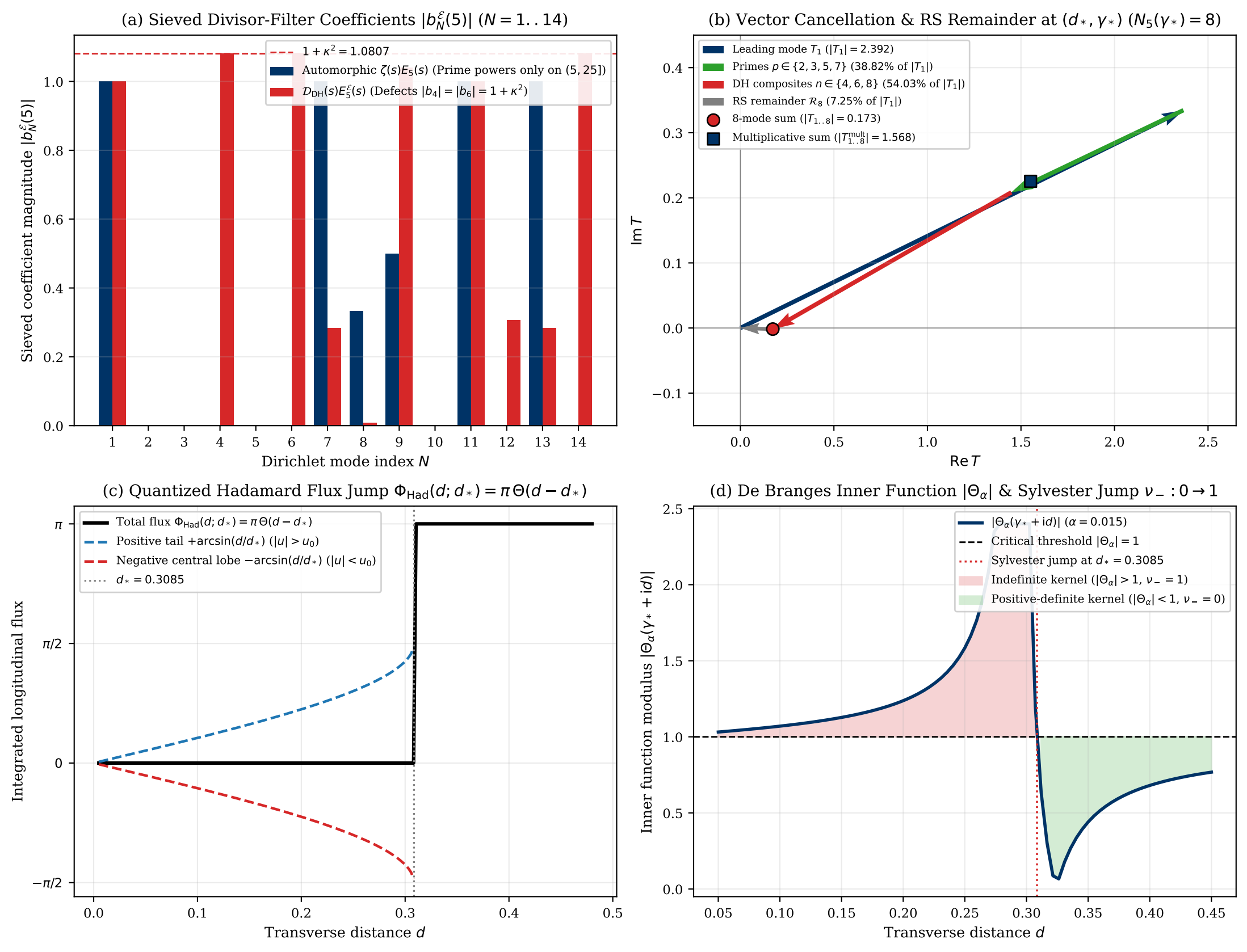

(a) Sieved Divisor-Filter Coefficients $|b_N^{\mathcal{E}}(5)|$ $(N = 1..14)$
$1 + \kappa^2 = 1.0807$
Automorphic $\zeta(s)E_5(s)$ (Prime powers only on (5, 25])
$D_{\mathrm{DH}}(s)E_5^{\mathcal{E}}(s)$ (Defects $|b_4| = |b_6| = 1 + \kappa^2$)
Sieved coefficient magnitude $|b_N^{\mathcal{E}}(5)|$
Dirichlet mode index N
(b) Vector Cancellation & RS Remainder at (d_*, γ_*) $(N_5(\gamma_*) = 8)$
Leading mode T_1 ($|T_1| = 2.392$)
Primes $p \in \{2, 3, 5, 7\}$ (38.82% of $|T_1|$)
DH composites $n \in \{4, 6, 8\}$ (54.03% of $|T_1|$)
RS remainder $\mathcal{R}_8$ (7.25% of $|T_1|$)
8-mode sum ($|T_{1..8}| = 0.173$)
Multiplicative sum ($|T_{1..8}^{\mathrm{mult}}| = 1.568$)
$\mathrm{Im}\, T$
$\mathrm{Re}\, T$
(c) Quantized Hadamard Flux Jump $\Phi_{\mathrm{Had}}(d; d_*) = \pi\Theta(d - d_*)$
Total flux $\Phi_{\mathrm{Had}}(d; d_*) = \pi\Theta(d - d_*)$
Positive tail $+\arcsin(d/d_*)$ ($|u| > u_0$)
Negative central lobe $-\arcsin(d/d_*)$ ($|u| < u_0$)
$d_* = 0.3085$
Integrated longitudinal flux
Transverse distance d
(d) De Branges Inner Function $|\Theta_\alpha|$ & Sylvester Jump $\nu_- : 0 \to 1$
$|\Theta_\alpha(\gamma_* + id)|$ ($\alpha = 0.015$)
Critical threshold $|\Theta_\alpha| = 1$
Sylvester jump at $d_* = 0.3085$
Indefinite kernel ($|\Theta_\alpha| > 1$, $\nu_- = 1$)
Positive-definite kernel ($|\Theta_\alpha| < 1$, $\nu_- = 0$)
Inner function modulus $|\Theta_\alpha(\gamma_* + id)|$
Transverse distance d